# A+: UNDERSTANDING OF NETWORK

## TASK: 1

## List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).



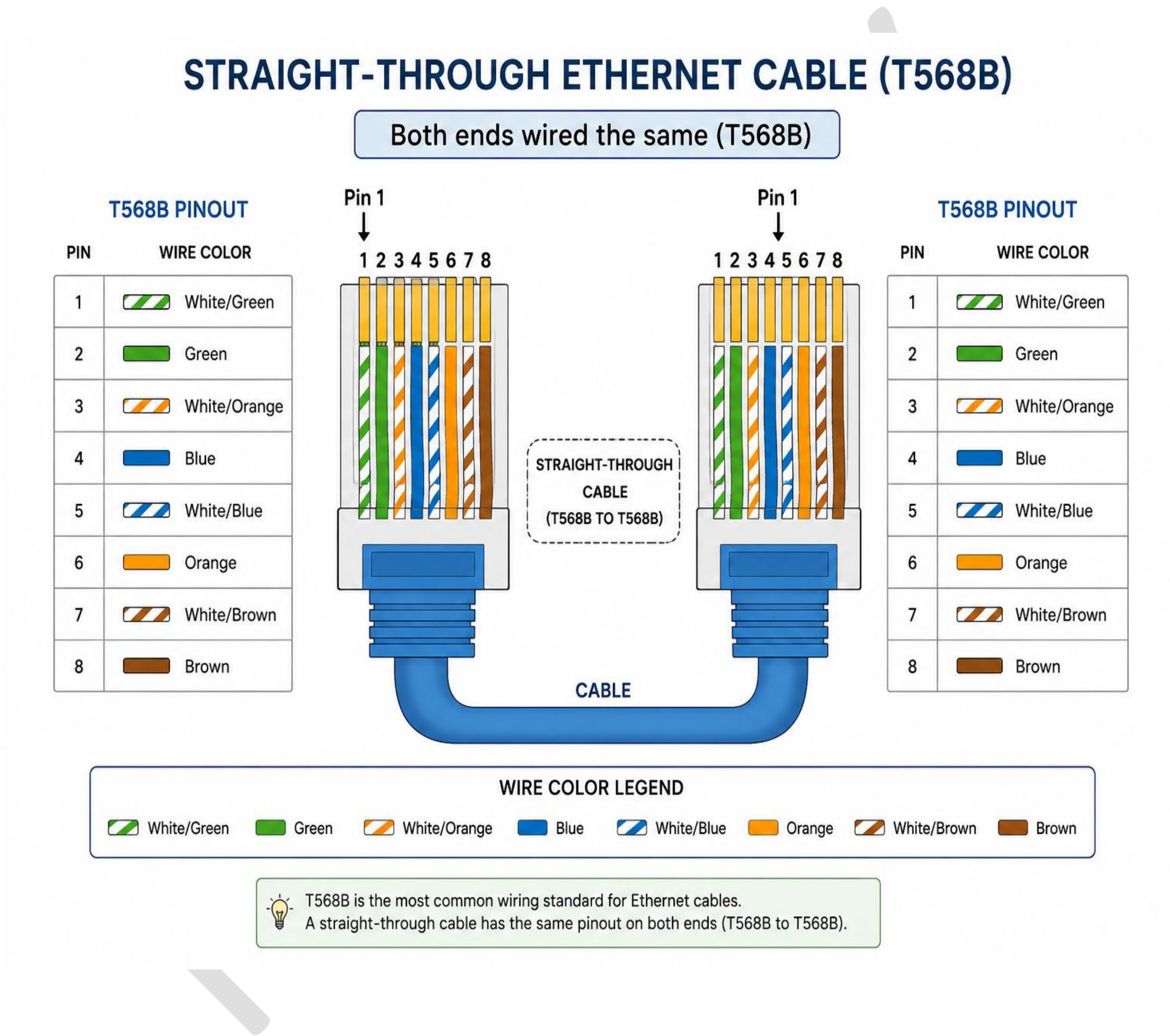
| Point of Difference | Ethernet Cable | Fiber Optic Cable |
|---|---|---|
| **Material & Signal** | Uses copper wires and transmits electrical signals. | Uses thin glass or plastic fibers and transmits light signals. |
| **Speed & Distance** | Generally suitable for shorter distances, commonly up to about 100 meters for standard Ethernet runs. | Supports very high speeds over much longer distances. |
| **Interference & Cost** | More affected by electromagnetic interference and usually cheaper/easier to install. | Resistant to electromagnetic interference but generally costs more to install. |

### Real-World Examples

- **Ethernet Cable:** Used to connect a **home Wi-Fi router to a desktop PC, laptop docking station, or gaming console**.
- **Fiber Optic Cable:** Used as a **college campus backbone or data center connection**, where high-speed communication over longer distances is required.

## TASK: 2

**Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.**



## TASK: 3

## Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

### Step 1: Cut the Cable

Cut the CAT6 cable to the required length using the cutter on the crimping tool.

### Step 2: Strip the Outer Jacket

Remove about **2–3 cm (around 1 inch)** of the outer plastic jacket carefully. Do not damage the eight small wires inside.

### Step 3: Separate and Straighten the Wires

Untwist the four pairs of wires and straighten all eight wires using your fingers. If the CAT6 cable has a plastic separator, cut it off carefully.

### Step 4: Arrange the Wires in T568B Order

Arrange the wires from **Pin 1 to Pin 8** in this exact order:

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown

### Step 5: Trim the Wires Evenly

Hold the wires tightly in the correct order and cut them straight so that all eight wires are the same length.

### Step 6: Insert the Wires into the RJ45 Connector

With the connector positioned correctly, carefully slide all eight wires into the RJ45 plug. Push them fully inside and make sure each wire reaches the end of the connector. The outer cable jacket should also enter the connector enough for proper strain relief.

### Step 7: Crimp the Connector

Insert the RJ45 connector into the correct slot of the crimping tool. Squeeze the handles firmly until the crimp is complete. This pushes the connector's metal contacts into the wires and secures the cable jacket.

### Step 8: Test the Cable

Connect both ends of the finished cable to a LAN cable tester. Check that pins **1 through 8** light up in the correct sequence. If all eight connections pass, the RJ45 connector has been attached successfully

**TASK: 4**

**Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.<br><br><em><strong>Hint:</strong> Consider speed, distance, and budget for your answer.</em>**

For a gaming café with 10 PCs connected for LAN gaming, I would choose **Ethernet cables**, preferably **Cat6**, because they provide high speed, low latency, and are affordable for short distances inside a café. Ethernet is much more budget-friendly than fiber optics and is easy to install between nearby gaming PCs and a network switch. I would use an **RJ45 crimper** to attach RJ45 connectors to the Ethernet cables. Fiber optics and a fiber cleaver would be more suitable for very long distances and high-speed backbone connections, but they would cost more than necessary for this setup.